

GRU initialization. The four GRU units in the interaction module (DF, RSSI $\times$ 2, and the Phase branch) adopt PyTorch default initialization. Setting of the initial hidden state: instead of zero initialization, we introduce four learnable query vectors, which read out node features via intra-modal self-attention, and the resulting representations are used as the initial hidden states of the respective GRUs. This enables the GRUs to possess modality-related priors from the very first iteration and avoids the gradient attenuation of zero initialization in the saturation region.

Attention head configuration. This paper uses three types of attention in total, with the head configurations as follows: (a) Feature extraction (GAT): both layers of graph attention convolution are configured with 2 attention heads, each with 128 dimensions. (b) Intra-modal self-attention: each modality is configured with 2 attention heads, each with 128 dimensions. (c) Cross-modal attention: a single-head (with a hidden dimension of the full 256 dimensions) gated attention is adopted: the query (source modality representation) and the key (target modality node features) are first projected through sigmoid gating, then a dimension-wise Hadamard product is computed, compressed into a scalar score via a linear layer with a applied sigmoid gate, and after softmax over the 16 nodes, the original node features are weighted and summed by using the weights.